

8.7. Prove that the series combination of the two tanks in Fig. 8.21(a) can be replaced with one tank as shown in Fig. 8.21(b).

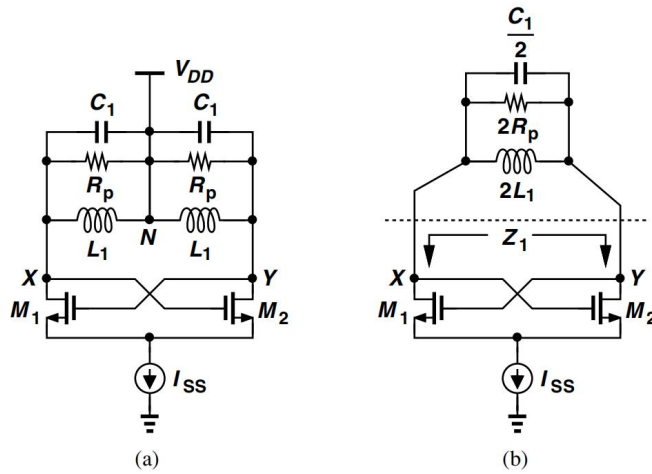


Figure 8.21 (a) Redrawing of cross-coupled oscillator, (b) load tanks merged

8.8. Compute the tuning range in Example 8.18 if C_b is placed at nodes P and Q in Fig. 8.32(a).

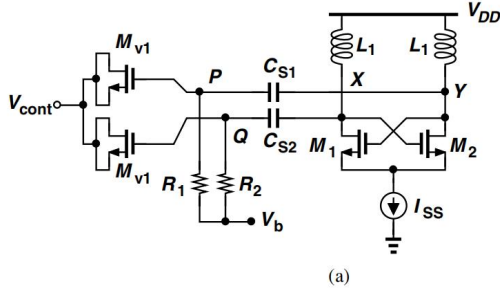


Figure 8.32 (a) VCO using capacitor coupling to varactors

Example 8.18

The VCO of Fig. 8.32(a) is designed for a tuning range of 10% without the series effect of C_S and parallel effect of C_b . If $C_S = 10C_{max}$, $C_{max} = 2C_{min}$, and $C_b = 0.05C_S$, determine the actual tuning range.

Solution:

Without the effects of C_S and C_b , Eq. (8.53) applies:

$$\Delta\omega_{osc} \approx \frac{1}{\sqrt{L_1 C_1}} \frac{0.5C_{max}}{2C_1}. \quad (8.72)$$

For this range to reach 10% of the center frequency, we have

$$C_{max} = \frac{2}{5}C_1. \quad (8.73)$$

With the effects of C_S and C_b , Eq. (8.71) yields

$$\Delta\omega_{osc} \approx \frac{1}{\sqrt{L_1(1.2C_1)}} \times \frac{0.43}{6} \quad (8.74)$$

$$\approx \frac{7.2\%}{\sqrt{1.2L_1 C_1}}. \quad (8.75)$$

The tuning range therefore falls to 7.2% (around $\sqrt{1.2L_1 C_1}^{-1}$).

$$\Delta\omega_{osc} \approx \frac{1}{\sqrt{L_1 C_1}} \frac{C_{var2} - C_{var1}}{2C_1}. \quad (8.53)$$

Let us now study the effect of the parasitics of C_{S1} and C_{S2} in Fig. 8.32(a). From Eq. (8.53), we note that a larger C_1 further limits the tuning range. In other words, the numerator of (8.53) *decreases* due to the series effect of C_S , and the denominator of (8.53) *increases* due to the parasitic capacitance of C_S . To formulate these limitations, we assume a typical case, $C_{max} \approx 2C_{min}$, and also $C_{var2} \approx C_{max}$, $C_{var1} \approx C_{min}$, $C_S = 10C_{max}$, and $C_b = 0.05C_S = 0.5C_{max}$. Equation (8.69) thus reduces to

$$\begin{aligned} \Delta\omega_{osc} &\approx \frac{1}{\sqrt{L_1(C_1 + 0.5C_{max})}} \times \frac{1}{2(C_1 + 0.5C_{max})} \times \\ &\times \frac{C_S^2(C_{max} - 0.5C_{max})}{(10C_{max} + C_{max})(10C_{max} + 0.5C_{max})} \end{aligned} \quad (8.70)$$

$$\approx \frac{1}{\sqrt{L_1(C_1 + 0.5C_{max})}} \times \frac{0.43C_{max}}{2(C_1 + 0.5C_{max})}. \quad (8.71)$$

8.13. Suppose the VCO of Fig. 8.25(a) employs varactors whose capacitance is given by $C_{var} = C_0(1 + \alpha_1 V + \alpha_2 V^2)$, where V denotes the gate-source voltage. Assume complete current steering and a low-frequency noise current of $I_n = I_m \cos \omega_m t$ in I_{SS} .

- Determine the AM noise resulting from I_n .
- Determine how the average value of the varactor capacitance varies with I_n .
- Compute the phase modulation as a result of the tank resonance frequency modulation.

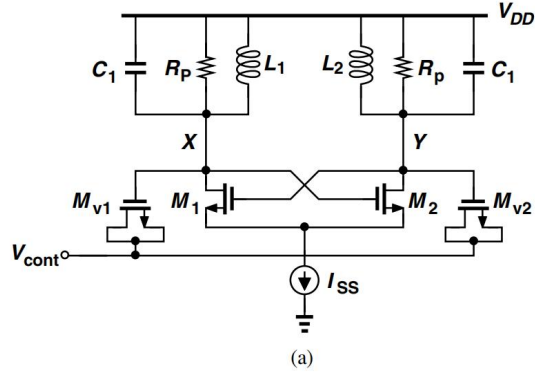


Figure 8.25 (a) VCO using MOS varactors

- 8.17. In the VCO circuit shown in Fig. 8.112, the voltage dependence of each varactor can be expressed as $C_{var} = C_0(1 + \alpha_1 V_{var})$, where V_{var} denotes the average voltage across the varactor. Use the narrowband FM approximation in this problem. Also, neglect all other capacitances and assume the circuit oscillates at a frequency of ω_0 for the given value of V_{cont} . The dc drop across the inductors is negligible.
- Compute the “gain” from I_{SS} to the output frequency, ω_{out} . That is, assume I_{SS} changes by a small value and calculate the voltage change across the varactors and hence the change in the output frequency.
 - Assume I_{SS} has a noise component that can be expressed as $I_n \cos \omega_n t$. Using the result found in (a), determine the frequency and relative magnitude of the resulting output sidebands of the oscillator.

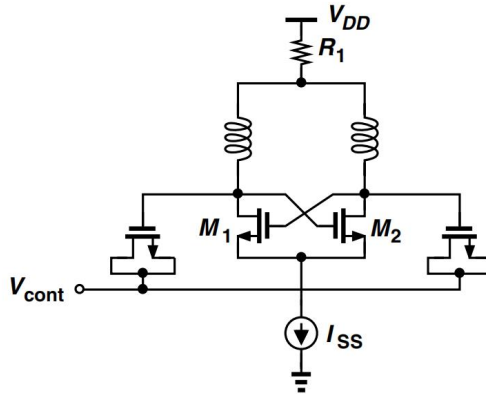


Figure 8.112 VCO with level-shift resistor.

- 8.18. The circuit shown in Fig. 8.113 is a simplified model of a “dual-mode” oscillator [28]. The voltage-dependent current source models a transistor. The circuit oscillates if Z_{in} goes to infinity for $s = j\omega$.
- Determine the input impedance Z_{in} .
 - Set the denominator of Z_{in} to zero for $s = j\omega$ and obtain the startup condition and the two oscillation frequencies.

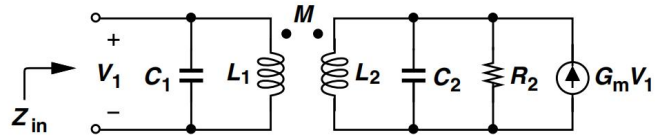


Figure 8.113 Simplified model of a dual-mode oscillator.